

DevOps Week 06 – Infrastructure as Code & Configuration Management

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1. Infrastructure as Code (IaC)

Infrastructure as Code is the practice of creating and managing infrastructure using code instead of manual configuration. It provides automation, consistency, and repeatability.

2. Terraform

Terraform is an Infrastructure as Code tool used to provision cloud infrastructure.

In this practical, Terraform was used to:

- Configure the AWS provider
- Create an EC2 instance
- Create a security group
- Configure SSH and HTTP access
- Use variables and outputs
- Create and destroy infrastructure

Important Commands

```
terraform init
terraform fmt
terraform validate
terraform plan
terraform apply
terraform destroy
```

3. Terraform State

Terraform State (`terraform.tfstate`) keeps track of the infrastructure managed by Terraform. Terraform uses it to determine what resources exist and what changes are required.

4. Variables and Outputs

Variables make Terraform configurations reusable by storing values such as AMI ID and instance type.

Outputs display useful information such as the EC2 instance ID and public IP address.

5. Modules

Terraform modules are reusable collections of Terraform configuration files. They help organize infrastructure and reduce repeated code.

6. Ansible

Ansible is an automation and configuration-management tool used to configure servers.

In this practical, Ansible was used to:

- Connect to the EC2 instance using SSH
- Install Nginx
- Create a `devops` user
- Start and enable the Nginx service

7. Ansible Inventory

The inventory file contains information about the servers managed by Ansible, including the server IP, username, and SSH key.

8. Ansible Playbook

A playbook is a YAML file containing automation tasks.

The playbook created in this practical installed Nginx, created the `devops` user, and started the Nginx service.

9. Terraform vs Ansible

Terraform	Ansible
Provisions infrastructure	Configures servers
Uses HCL	Uses YAML
Creates EC2, networks, etc.	Installs software and manages services
Infrastructure as Code	Configuration Management

Simple difference: Terraform creates the server, while Ansible configures the server.

10. Practical Workflow

Terraform



AWS EC2



Ansible



Install Nginx + Create User



Nginx Web Server

The implementation was successfully completed, and the Nginx welcome page was verified in the browser.

11. Conclusion

This practical provided hands-on experience with Terraform and Ansible. Terraform was used to provision AWS infrastructure, while Ansible automated the configuration of the EC2 server. Together, they demonstrated a basic DevOps infrastructure automation workflow.